

AIRWISE

India Air Quality Intelligence Dashboard

Professional Project Report

Framework	Streamlit
Data Processing	Pandas, NumPy
Visualization	Plotly Express, Matplotlib
Dataset	Air_quality_in_india.csv (3,514 records)
Report date	September 2026

Prepared for IBM Internship / GitHub Portfolio

All figures in this report are computed directly from the supplied dataset and application code.

Table of Contents

1. Introduction
2. Objectives
3. Dataset Overview
4. Data Cleaning & Quality Checks
5. Dashboard Features
6. Methodology
7. Exploratory Data Analysis & Findings
8. Technical Implementation
9. Limitations
10. Future Scope
11. Conclusion

1. Introduction

AIRWISE — India Air Quality Intelligence Dashboard is a Streamlit-based analytics application built to make air-quality observations easy to explore at the national, state, city, station and pollutant level. It combines data cleaning, descriptive analysis, interactive visualization, geographic exploration and automatically generated data-driven insights in a single, self-contained app.

The application implements five main sections — Overview, Map, Pollutant Analysis, State & City Comparison, and Insights — built on top of a cleaned and validated version of the supplied dataset.

2. Objectives

- Inspect, clean and validate the supplied air-quality dataset.
- Provide a clear, KPI-driven overview of pollutant levels across India.
- Enable geographic exploration using station latitude/longitude data.
- Compare pollutant levels across states, cities and monitoring stations.
- Generate insights directly from the observed dataset, with no fabricated forecasts.
- Present the analysis through a robust, attractive, and genuinely interactive interface.

3. Dataset Overview

The application reads **Air_quality_in_india.csv**. Key structural facts, computed directly from the file:

Metric	Observed Value
Raw rows	3,514
Raw columns	11
States represented	30
Cities represented	261
Monitoring stations	502
Pollutant types	7 (CO, NH3, NO2, OZONE, PM10, PM2.5, SO2)
Timestamp coverage	22-09-2026 22:00:00 (a single real-time snapshot, not a time series)
Duplicate rows	0

Core fields: *country, state, city, station, last_update, latitude, longitude, pollutant_id, pollutant_min, pollutant_max, pollutant_avg*.

4. Data Cleaning & Quality Checks

4.1 Missing values

The raw file contains 990 missing cells, concentrated entirely in the three pollutant measure columns. The application imputes these using the column median, computed after filtering.

Column	Missing cells	Treatment
pollutant_min	330	Median imputation
pollutant_max	330	Median imputation
pollutant_avg	330	Median imputation

4.2 Other checks

- Duplicate rows: 0 fully duplicated rows found; the app does not currently drop duplicates as a result, since none were present to remove.
- Timestamp parsing: *last_update* is parsed as DD-MM-YYYY HH:MM:SS with invalid values coerced to missing and dropped; 0 rows failed to parse.
- Text fields (state, city, station, pollutant_id) are stripped of stray leading/trailing whitespace as part of the current cleaning step, preventing duplicate categories caused by formatting alone.

5. Dashboard Features

The dashboard is organised into five interactive tabs, each covering a distinct layer of analysis from national KPIs down to individual monitoring stations.

Tab	Purpose
Overview	KPI summary cards, a top-15 station ranking for the selected pollutant, and a category breakdown donut chart.
Map	Geographic scatter map of stations, colour-switchable between raw concentration and AQI-style category.
Pollutant Analysis	Grouped bar chart comparing pollutants by state, and a box plot of each pollutant's distribution.
State & City Comparison	Top/bottom-10 city rankings, full state ranking, an inline city search box, and a filtered-data CSV download.
Insights	Automatically generated, deterministic textual insights computed from the currently filtered data — not a machine-learning forecast.

All pollutant readings are additionally classified into a CPCB-style category (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) for PM2.5 and PM10, where the standard breakpoints are well-established for the dataset's microgram-per-cubic-metre units. Other pollutants are labelled with a dataset-relative Low/Moderate/High/Very High scale, since their exact reporting units are not confirmed in the source file.

Global filters in the sidebar let a user scope the whole dashboard to All India, a chosen set of states, or a chosen set of cities, with the pollutant selector driving the Map, Comparison and Insights tabs.

6. Methodology

The workflow follows a practical exploratory-data-analysis pipeline: load the CSV (with a resilient file-matching strategy), clean text fields, parse the timestamp, impute missing pollutant values, classify readings into AQI-style categories, aggregate by geography and pollutant, and present results through interactive Streamlit + Plotly components. Streamlit handles the application layer and state; Pandas performs the tabular processing; NumPy backs numeric operations; Plotly Express renders the interactive

charts and map.

7. Exploratory Data Analysis & Findings

The following are descriptive statistics computed directly from the cleaned dataset. They characterise this snapshot and dataset only, not a comprehensive assessment of India's air quality.

Pollutant	Obs.	Mean	Median	Min	Max
PM10	502	76.86	63.50	12.00	342.00
PM2.5	502	68.26	49.50	2.00	405.00
CO	502	37.26	31.00	2.00	195.00
OZONE	502	31.50	26.00	0.00	268.00
NO2	502	23.51	21.00	0.00	163.00
SO2	502	14.58	12.00	0.00	108.00
NH3	502	8.53	5.00	0.00	31.00

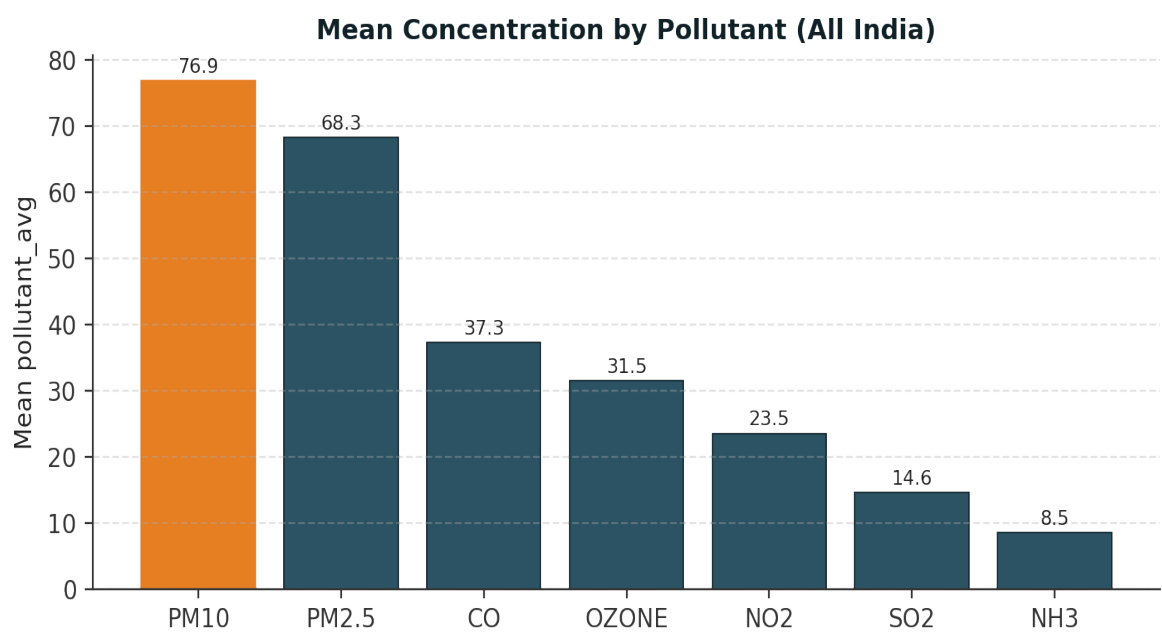


Figure 1 — Mean pollutant_avg across all stations, by pollutant type.

Across all pollutant observations combined, the state-level mean of pollutant_avg is highest for **Ladakh (81.14)** and lowest for **Sikkim (6.14)**. Because different pollutants are measured on different scales, this combined figure should not be read as a universal air-quality ranking.

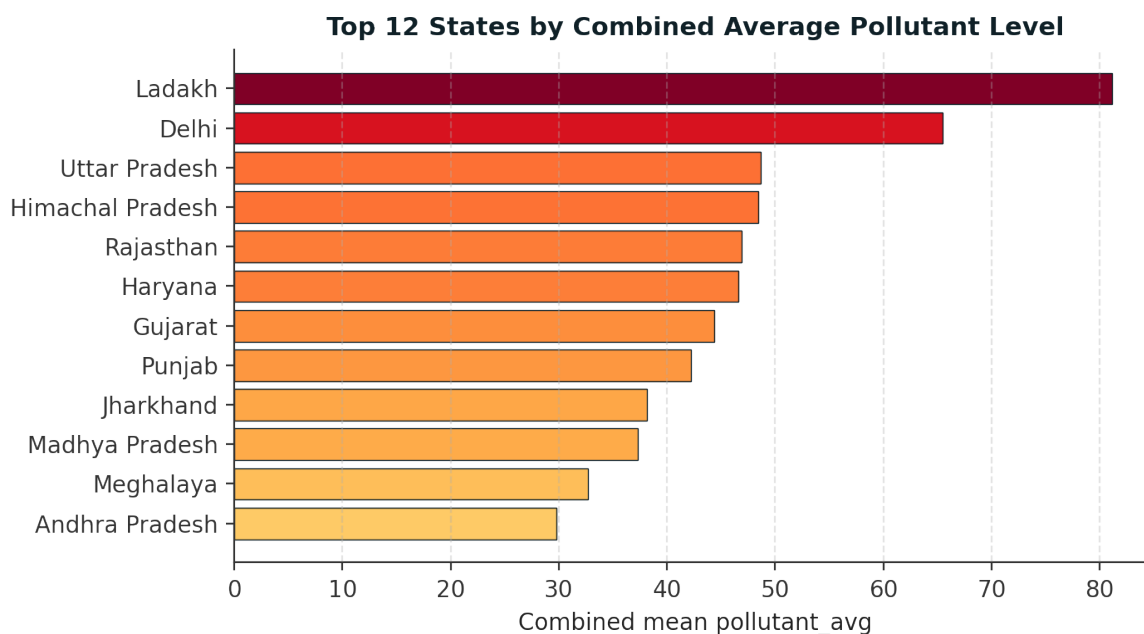


Figure 2 — Top 12 states by combined mean pollutant_avg across all pollutant types.

At city level, the highest combined mean pollutant_avg is observed for **Khora (101.86)**, while the lowest is **Gangtok (6.14)**. Using the CPCB PM2.5 breakpoints specifically, of the 502 PM2.5 station readings in this snapshot, 137 fall in the Good band, 173 Satisfactory, 78 Moderate, 30 Poor, 82 Very Poor and 2 Severe.

PM2.5 Station Readings by Category

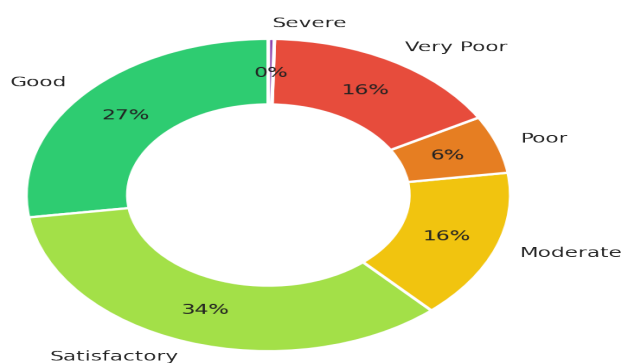


Figure 3 — Share of PM2.5 station readings by CPCB-style category, this snapshot.

8. Technical Implementation

The Streamlit application is contained in a single `app.py` file. Data loading and cleaning are cached with Streamlit's `@st.cache_data` mechanism so repeated filter interactions do not re-read or re-clean the CSV. Global state, city and pollutant filters are resolved before any analytical tab renders; each tab then filters the already-cleaned DataFrame in place.

- Plotly renders the station ranking bars, category donut, geographic map, grouped bar charts and box plots — all filter-responsive and hoverable.
- The map switches between a continuous colour scale (raw concentration) and a discrete, CPCB-coloured scale (category) via a radio control.
- A CSV export of the currently filtered data is available directly from the Comparison tab.

9. Limitations

- The dataset is a single real-time snapshot; it does not support genuine trend-over-time analysis, only a cross-sectional view of one moment.
- Pollutant values may use different measurement units, so cross-pollutant numeric comparisons require care.
- Median imputation for missing pollutant measures may understate the true variability of those fields.
- Non-PM pollutant categories (CO, NO2, SO2, OZONE, NH3) use dataset-relative quartile labels rather than official CPCB AQI sub-index calculations, since exact reporting units are not confirmed.
- Correlation, causal inference and health-risk assessment remain outside the scope of this dashboard.

10. Future Scope

- A formal, always-visible data-quality panel (missing %, duplicate rows, parsing failures).
- Pollutant-specific unit confirmation and full CPCB AQI sub-index calculation for all seven pollutants.
- Historical ingestion so genuine time-series trend charts become meaningful.
- Station-level drill-down pages and anomaly detection for outlier readings.
- Validated forecasting models, clearly separated from the current descriptive-only insights.

11. Conclusion

AIRWISE provides a portfolio-ready framework for exploring India's air-quality observations through an interactive Streamlit interface. Its value comes from connecting careful data preparation with intuitive geographic, comparative and category-based views, while keeping every generated insight directly traceable to the underlying dataset.